The coefficient of variation

How much the replicates of a condition disagree about the same feature, in per cent of their mean.

AT A GLANCE

Where	Quality control → Coefficient of variation
Data levels	Protein, phosphosite, phosphoprotein
Needs	An uploaded dataset and its annotation, and at least one condition with two or more samples.
Data source	The normalized, unfiltered, non-imputed QC matrix, converted back to the linear scale for this calculation; after VSN, the unnormalized data.
Varies across	Replicates of one condition, for one feature. Not across features, and not across conditions.
One box	One condition. Its values are the CVs of all features quantified in at least two of that condition's replicates.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

WHEN NORMALIZATION IS VSN

The calculation exponentiates the analysis values to obtain positive values for a CV. VSN output is log-like rather than an ordinary logarithm, so exponentiating it does not give a linear intensity. After VSN normalization this page therefore uses the unnormalized data of the same rows, and a note under the plot says so: VSN output is not on a linear intensity scale, so this plot uses the unnormalized data. With any other

> normalization the normalized values are used.

1 What the plot answers

The coefficient of variation asks how far apart the replicates of a condition are when they measure the same thing, as a fraction of what they measured. Two conditions can have different mean intensities and similar relative variability. Read the median and the width of the box together. A low CV describes consistent values under the chosen processing; it does not establish accuracy, independent biological replication or suitability for every downstream test.

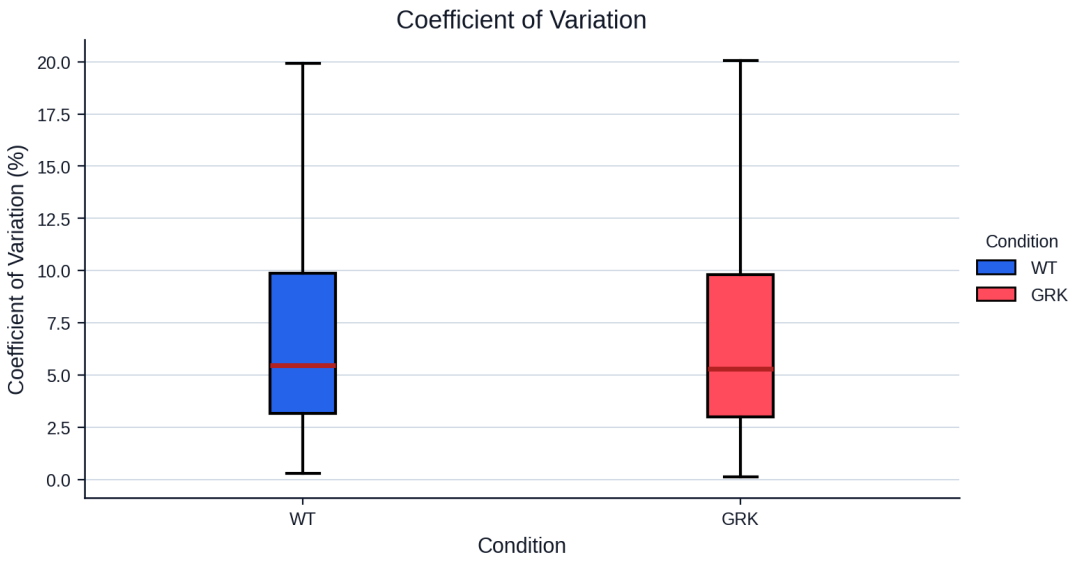


Figure 1.1. Protein level on the example dataset: one box for the five WT replicates and one for the five GRK replicates. Both medians sit near 5 % and both boxes close below 10 %. These are distributions of per-feature relative standard deviations, not confidence intervals or error bounds for each measurement.

Reading it

- One box, one condition, built from one CV per feature.
- The median line: the CV of the typical feature in that condition. With **Toggle Text** on its value is printed at the line, to two decimals.
- The upper whisker and the outliers above it: the features whose replicates disagreed most. A few are expected — low-abundance features have poor precision — but a long tail common to every condition is a property of the run.
- The y axis is per cent of the mean, so the boxes of different conditions are directly comparable even when their intensities are not.

EASY TO GET WRONG

Compute an intensity CV on a positive linear scale, not by dividing the standard deviation of log intensities by their mean. The problem is the origin: multiplying linear intensities by a unit-conversion factor adds a constant to their logarithms, changing that log-scale denominator. Changing only the logarithm base scales both numerator and denominator equally; it is not the reason the log-scale calculation is unsuitable. The notebook function requires an explicit linear-scale declaration.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw: protein, phosphosite or phosphoprotein. Only the levels actually uploaded are offered; phosphoprotein appears once a collapse has been run.	Protein	page
Toggle Outliers	Draws the points beyond the whiskers. With it off the whiskers are still computed the same way; the points outside them are simply not marked.	off	page
Toggle Text	Prints the median CV at each box, to two decimals.	off	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, DPI between 72 and 600.	20 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Box border color	The outline of every box, and of its whiskers and caps.	black	appearance
Median color	The line across the box.	firebrick	appearance

In the plot appearance panel (continued)

Control	What it does	Default	Where
Outlier dot color, Outlier outline color, Outlier size (pt)	The markers beyond the whiskers. The three rows appear only while **Toggle Outliers** is on.	red, black, 5	appearance
Value labels	Size of the number printed at each box by **Toggle Text**. Until it is set the plot uses 9.	9	appearance
Condition names, Condition colors	One display name and one colour per condition; the colour is the box fill.	the condition palette	appearance
Title	Shown or hidden, and its text, size, weight and colour. The title is written by the plot: Coefficient of Variation.	shown	appearance
Legend	Shown or hidden, and its title, size, colour, frame and position. Its entries are the conditions.	shown	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

The box, the whiskers and what counts as an outlier follow the standard definition given in **Concepts**; it is not repeated here. The box on this page contains one CV per feature, and the CVs of one condition only.

HOW IT IS COMPUTED

1. Use the normalized QC matrix, before completeness filtering and imputation — after VSN, the unnormalized values of the same rows — and convert its stored values to linear values. Missing entries remain missing. The result describes variation after the selected normalization, not raw instrument precision.
2. Conditions with fewer than two samples are dropped. A single replicate has nothing to vary across, and such a condition gets no box and no place on the axis.

3. Within each remaining condition c, and for each feature i, the mean and the sample standard deviation are taken across that condition's replicates, over the observed values only:

$$m_{i,c} = \frac{1}{n_{i,c}} \sum_{j \in O_{i,c}} x_{ij}, \qquad s_{i,c} = \sqrt{\frac{1}{n_{i,c} - 1} \sum_{j \in O_{i,c}} \left(x_{ij} - m_{i,c}\right)^2},$$

   where $O_{i,c}$ are the replicates of c in which feature i was quantified and $n_{i,c} = |O_{i,c}|$. The denominator is $n - 1$, so a feature seen in only one replicate of the condition has no standard deviation.

4. The coefficient of variation, in per cent:

$$\mathrm{CV}_{i,c} = \frac{s_{i,c}}{m_{i,c}} \times 100.$$

5. Every value that is not finite is removed — the features seen in one replicate, and any whose mean is zero. What remains is the vector the box of condition c is built from.

> **NOTE**
>
> Note what the CV is *not* taken over. It is not the spread of one sample's intensities across features, which is a property of the dynamic range and not of reproducibility; and it is not the spread of a feature across conditions, which is the biological effect the experiment is looking for. It varies across replicates within one condition, one feature at a time.

For values 80, 100 and 120, the mean is 100 and the sample standard deviation is 20, giving a CV of 20 %. Multiplying all three by ten leaves the CV unchanged. With only two observed replicates a CV can be calculated, but it is an uncertain estimate.

A higher box means greater relative variation among the observed values. Possible explanations include biological heterogeneity, low signal and technical variability; the box alone cannot separate them. Inspect sample coverage, intensity distributions and the individual features before making a quality decision. There is no universal acceptable CV threshold for all proteins, experimental designs and abundance ranges.

Edge cases

- A condition with one replicate has no box. It is not drawn empty and it is not drawn at zero: it is absent from the axis, and from the table.
- A feature quantified in only one replicate of a condition contributes no CV to that condition, but may contribute one to another condition where it was seen twice. Boxes therefore rest on different numbers of features.
- If no condition has two samples, no figure is drawn and the page reports that instead.

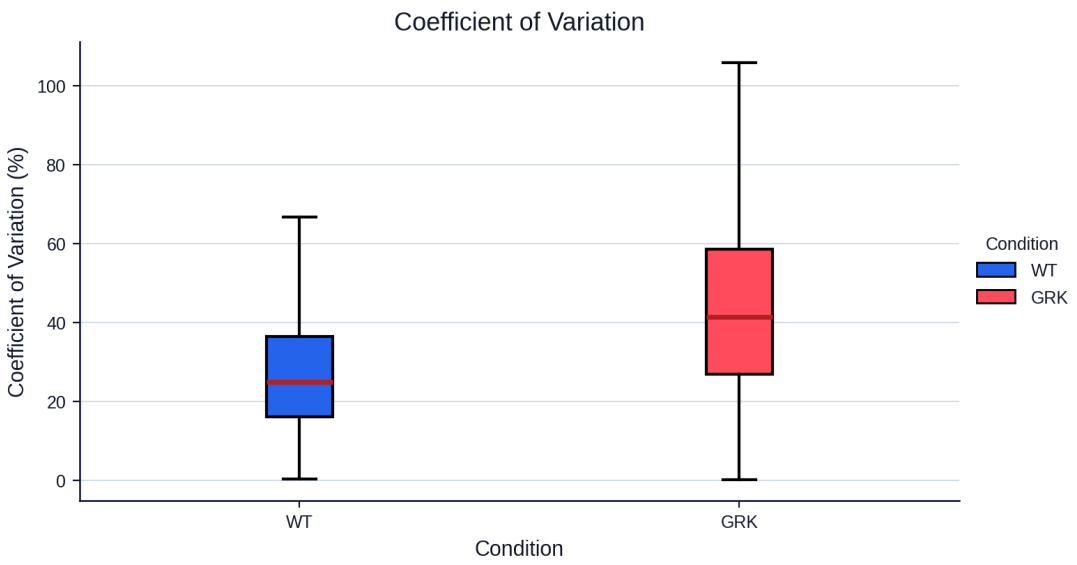


Figure 1.2. The same experiment at phosphosite level, and a different axis: the scale now runs past 100 %, the WT median sits near 25 % and the GRK box is both higher and much wider. Enrichment costs precision, and the comparison worth making is between the two boxes here rather than between either of them and the protein-level figure above.

- Features whose replicates agree exactly give a CV of zero, which is a legitimate value and is kept.
- At phosphoprotein level collapsed with `robust_site_median` the stored values are ratios to each site's own median, so on the linear scale they are centred on one rather than on an abundance. A ratio to the standard deviation of such values is arithmetically defined but is not a coefficient of variation of an abundance, and should not be compared with the protein-level numbers. The page draws them anyway, and prints a warning under the plot when it does, because the number is defined but is not the quantity the page is named after.

4 The table below the plot

One row per condition that has a box, holding the numbers that draw it, each rounded to two decimals.

Columns in this table

Column	What it holds
Condition	The condition, as the annotation names it. Conditions with fewer than two samples do not appear.

Columns in this table (continued)

Column	What it holds
mean	The arithmetic mean of that condition's CVs. It is not drawn on the plot, and it is not the median.
median	The line across the box — the CV of the typical feature.
Q1	The lower edge of the box, the 25th percentile of the CVs.
Q3	The upper edge of the box, the 75th percentile of the CVs.
lower_whisker	The smallest CV at or above $Q1 - 1.5 \times \text{IQR}$.
upper_whisker	The largest CV at or below $Q3 + 1.5 \times \text{IQR}$ — the most useful single number on the table for comparing conditions.

Every column is in per cent, and the two whisker columns are observed CVs rather than the cutoffs themselves.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = cov_plot(org_data, meta, data_scale="linear")
```

R

```
fig <- cov_plot(org_data, meta, data_scale = "linear")
```

The frame is org_data, the linear one obtained with inverse_log2_transform_data, because a CV on $\log_2$ values is not a CV; and data_scale is a declaration rather than a conversion — the function refuses to run without it instead of turning a $\log_2$ matrix into a plausible-looking number. outliers=True and text=True are **Toggle Outliers** and **Toggle Text**.

6 Exporting

The download reproduces the plot at the width, height and resolution set on the page. The browser scales the rendered figure to fit the page. Choose PDF for a figure that will be scaled or edited, PNG for a slide. The table underneath downloads separately as CSV.

7 Further reading

NIST: coefficient of variation explains the ratio of standard deviation to mean and the importance of a meaningful zero.

Behind this page

Function	What it produces
cov_plot	One box of CVs per condition. It requires the caller to state that the matrix is linear, and refuses a $\log_2$ one
prepare_cov_df	The table under the plot